

Week 1: Jumping-peg games

Peg solitaire on strips and small boards, and Conway's soldiers

Facilitator guide for the three table adults. Student pages: *week-k-1* (K–1, 12 pages), *week-grades-2-3* (19 pages), *week-grades-4-5* (21 pages).

1. The session at a glance

0:00–0:05	Free handling. Strips and pieces are already on the tables; children play with them. The adult only answers questions.
0:05–0:08	Whole-group launch (Section 4), at the K–1 table where the pieces are biggest.
0:08–0:50	Table work (Sections 5–7). Each table has a core sequence of about forty minutes and reserve problems for a fast table: the K–1 core ends with Problem 9, the grades 2–3 core with Problem 14, the grades 4–5 core with Problem 10; everything printed after that is reserve, in the printed order.
0:50–0:58	Share (Section 8). Each table shows one thing; the organizer closes with the theorem about the fifth row.
0:58–1:00	Pieces back into cups, packets into folders.

Who sits where. The K–1 section of this guide is a complete read-aloud script and needs no mathematics beyond counting, so it is the natural table for the parent; the two mathematicians take grades 2–3 and 4–5. If the organizer prefers to take K–1, swap: every table's section is complete on its own. Whoever sits with grades 4–5 should read Section 9 beforehand and play the twenty-soldier sequence (Problem 20 of the 4–5 pages) once at home; it takes five minutes.

The one rule of the day. A piece jumps in a straight line (left, right, up or down, never diagonally) over a piece in the next square into the empty square immediately beyond, and the piece jumped over is removed. That is the only move. “Finished” means one piece left, anywhere. “Where does the last piece end” is a separate follow-up question.

The questions that run through every table. Can this board be reduced to one piece? When it cannot, how can we be sure without trying everything? What never changes during a game (an invariant)? What is the fewest pieces we can leave, or the fewest soldiers that get across? Every table meets the same four questions at its own size; Section 9 says what they are called.

2. Materials and preparation

- **Printing**, single-sided so boards lie flat. K–1 packet: 4 copies (three children and a spare), plus 2 extra copies of pages 2–4 (the strips get marked up). Grades 2–3 packet: 5 copies (four children and a spare), plus 2 extra copies of pages 1–2. Grades 4–5 packet: 4 copies, plus 2 extra copies of page 1. One extra copy of K–1 pages 1 and 2 for the launch. This guide: 3 copies, one per adult.
- **Checkers** for the K–1 table: about 12 per child, 36 in all (the largest K–1 board needs 8 and the reserve strips use pennies). Two checkers sets give 48. Standard checkers are 1.25 in across; the K–1 cells are 1.4 in (1.3 in on the square boards), so they will not overlap.
- **Pennies** for the other two tables: about 20 per child at grades 2–3 (80) and 25 per child at grades 4–5 (75: the twenty-soldier army needs twenty and attempts with sixteen or more need spares), plus 8 per K–1 child for the reserve strip. About 180 pennies, sorted in advance into ten cups of 18–25.

Pennies roll; every cell has a printed dot to sit them on. A penny is 0.75 in across and fits every cell of 0.78 in and up, which is every cell a penny is placed on in the packets except the two prediction strips of grades 4–5 Problem 14 (0.7 in): put a cup of the smaller counters (0.5 in), or dimes, on that table for those two strips.

- **Ten small cups or bowls**, one per child, for removed pieces. The cup is the jump counter (K–1 Problem 5 and its follow-ups): pieces at the start = pieces in the cup + pieces on the board, always.
- **Crayons or coloured pencils** in red, blue and green: two sets at the grades 2–3 table, one or two at grades 4–5. Any crayons at K–1 (for the small army boards of Problems 8, 9 and 11 and the farthest square of Problem 12; the circles and X marks are drawn in pencil).
- **Pencils** for everyone, a few erasers, scrap paper at the two older tables.
- Before the children arrive, put on each table the packets, a cup of pieces per child, and one open packet turned to page 1 for the free-handling minutes.

3. The rule, and how it goes wrong

State the rule once, with the pieces, and then only correct play, never explain: “hop over a neighbour into the empty square right behind it, and take the neighbour off.” The mistakes to watch for, in order of frequency:

- sliding a piece into an empty square next to it (no: a move is always a jump);
- jumping over an empty square, or over two pieces (no: exactly one neighbour, and the landing square must be the very next one);
- jumping diagonally on the square boards (no);
- jumping off the end of a strip or board (the printed board is the whole world);
- forgetting to remove the piece that was jumped over (this is the mistake that produces impossible “solutions”);
- a double jump made as one move (harmless, but the child then miscounts jumps; count it as two);
- for soldiers: placing a soldier above the line at the start, or asking whether “the same” soldier must reach the star (any soldier on the star wins; identity is irrelevant).

If a child reports a result that contradicts this guide (“the five-strip can be finished”, “the corner hole on the three-by-three gets down to three”, “sixteen soldiers reach row 4”), one of the mistakes above was made; ask the child to replay it slowly while you watch the removed pieces.

Mirror images: the hole on square 2 and the hole on square 3 of the four-strip are the same problem seen from the other end. Children may treat them as different cases; that is fine, and noticing the symmetry is itself a discovery worth a “so which other holes do you already know the answer for?”

4. The whole-group launch (three minutes)

Gather everyone around the K–1 table, where the strips are large. Use the extra copy of K–1 pages 1 and 2 with checkers.

1. Page 1, top strip (checkers on 1 and 2, square 3 empty). “One move in today’s game.” Ask a child to hop the checker on 1 over the checker on 2 into square 3, and take the jumped checker off. Repeat the rule in one sentence: hop over a neighbour into the empty square right behind it, and take the neighbour off.
2. Page 1, bottom strip (checkers on 1 and 3, the middle empty). “Can anyone move?” No: there is nothing next to a checker to hop over. A single checker with a gap can never jump.
3. Page 2, top strip (four squares, the hole at the end). Ask a child to play it out: one jump ($3>2>1$), and then the two checkers left, on squares 1 and 4, are not neighbours. Stuck with two.
4. “The empty square is in the wrong place. At your table, find where it has to be. At every table today you will ask: which boards can be cleared down to one piece? When you cannot, how can you be sure? And later, an army of pieces will try to cross a line: how far can it get?” Send the tables off. Do not play the hole-on-2 strip and do not explain any answer; the first problem at every table is to find the hole that works.

5. The K–1 table (two kindergartners, one first grader)

How the table runs. You read every problem aloud, one sentence at a time, and the children do it with checkers. Nobody needs to read or write; the record is a circle drawn where a checker stood, an X on a chosen square, or a number you write for them. Give each child their own packet and cup; let them work side by side on the same problem and compare. When a child is stuck, say less rather than more: “show me every jump you could make from here” is usually enough.

Timing (about 36 minutes). Problem 1: 2 minutes (played in the launch; let them redo it with their own hands). Problem 2: 4. Problem 3: 5. Problem 4: 5. Problem 5: 4. Problem 6: 3. Problem 7: 5. Problem 8: 2. Problem 9: 6. The core ends with Problem 9 (the star two rows up). Reserves: Problems 10, 11, 12 (about 2, 7 and 5 minutes). A child who finishes a board before the others answers the cup question on it (“how many jumps did you make? count the cup”) and then tries to make the last checker end somewhere else.

Running questions for every board (ask them each time; they are the mathematics of this table): “Show me every jump you can make from here.” “Why can’t those two jump?” (they are not next to each other, or there is no empty square behind). “Before you play: if you get down to one checker, how many will be in the cup?” (one less than you started with). “We tried the hole in every place; which ones worked?”

Problem 1 (two strips of three)

Top strip: the only jump is $1>2>3$; one checker left, on square 3; one circle. Bottom strip (checkers on 1 and 3): no jump is possible; two circles. Say back what they found: a checker needs a neighbour to hop over *and* an empty square right behind the neighbour.

Problem 2 (four strips of four, the hole in every place)

Hole on 1: only $3>2>1$ is possible; then the checkers on 1 and 4 are not next to each other; stuck with two. Hole on 2: $4>3>2$, then $1>2>3$; one left, on square 3. Hole on 3: the mirror image, $1>2>3$ then $4>3>2$; one

left, on square 2. Hole on 4: only $2>3>4$; stuck with two (on 1 and 4). The strips with the hole *next to* an end can be finished; the strips with the hole *at* an end cannot. Hint if a child does not see the second jump on the hole-2 strip: “which checker has a neighbour with an empty square behind it now?” Ask why the end-hole strips got stuck: “after your jump, are the two left next to each other?” That sentence, said by a five-year-old, is the whole proof, because there was only one possible first jump.

Problem 3 (five strips of five)

No strip of five can be finished; every strip ends with exactly two checkers, whatever is played. The complete list of games (there is almost no choice): hole 1: $3>2>1$, $5>4>3$, stuck with checkers on 1 and 3. Hole 2: $4>3>2$, $1>2>3$, stuck on 3 and 5. Hole 3: either $1>2>3$ then $4>3>2$ (stuck on 2 and 5) or the mirror $5>4>3$ then $2>3>4$ (stuck on 1 and 4). Holes 4 and 5 are the mirror images of holes 2 and 1. Expected wrong report: “I finished it” (a jump over an empty square, or a jumped checker left on the strip); replay it. The argument the children can give: “we tried every hole, and there was never a real choice, and it always ended with two.” Point at the circles: each strip shows two circles with a gap between them.

Problem 4 (strips of six, choose the empty square)

Before they start, share out the X squares so that the table covers every square: one child tries 1, 2, 3 from the top, another 4, 5, 6, the third 2, 3, 4; then pool: “which X squares worked?” The X must be on square 2 or square 5 (counting from the top). Then four jumps finish the strip; from the hole on 2: $4>3>2$, $1>2>3$, $6>5>4$, and then a real choice, $3>4>5$ (the last checker ends on 5) or $4>3>2$ (it ends on 2). Hole on 1 or 6: three left at best ($3>2>1$, $5>4>3$; stuck on 1, 3, 6). Hole on 3 or 4: two left at best ($1>2>3$, $4>3>2$, $6>5>4$; stuck on 2 and 4). Hint: “on the strips of four, where was the hole when it worked?” (next to the end). Second hint: “start with the hole on the second square and look for a jump into it.” A child who finishes it should be asked to do it again and make the last checker end on the other square.

Problem 5 (the cup)

Top strip (four squares, hole on 2): start 3, cup 2, strip 1. Bottom strip (five squares, hole on 2): start 4, cup 2, strip 2. In both, start = cup + strip. You write the numbers for them. Why: every jump puts exactly one checker in the cup, no more and no less, so the cup counts the jumps. The prediction before the bottom strip: “if you end with one, the cup will have 3” (start 4, one left). It does not happen, the strip gets stuck with two and the cup holds two, and that is the point: the number of jumps in a finished game is known before playing (one less than the start), and a game that stops early has fewer. Use this on every later board: “how many jumps did you make? Count the cup.”

Problem 6 (three-by-three: hole in the centre, hole in the middle of the top edge)

Top board (hole in the centre): no jump exists; the box gets 0 and there are eight circles. Every line of three that could be jumped along either is full (the edges) or has the empty square in the middle (through the centre), and a jump needs its empty square at an end. Let the children try every checker; “show me a jump” produces none, and “none” is the right answer. Say it back: “the empty square is in the middle of every line that goes through it.” Bottom board (hole in the middle of the top edge): exactly one jump, the

bottom-middle checker over the centre into the top-middle square; the box gets 1 and there are seven circles. After it, the centre and the bottom-middle square are empty and no line of three has a checker, a checker, and an empty square in a row. Ask “why can nothing move now?” and accept a pointed-at answer: every line either is full or has a gap in the middle.

Problem 7 (three-by-three, hole in a corner; two boards)

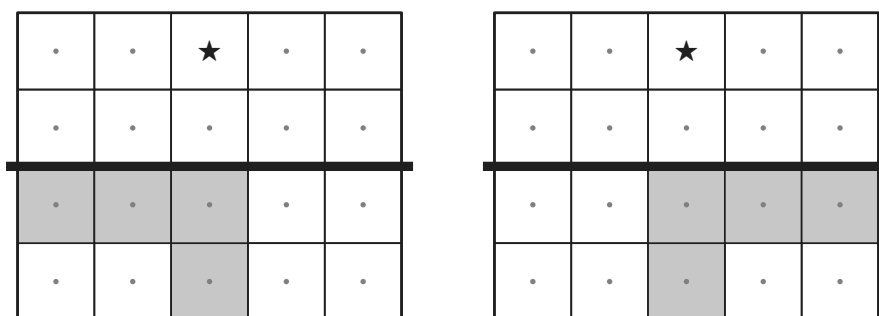
The fewest checkers that can be left is four; three is impossible. There are only fourteen ways to play to the end: ten of them end with four checkers, four with five. A line to four, with the squares called A B C (top row), D E F, G H I: $C > B > A$, $H > E > B$, $A > B > C$, $G > D > A$; the four left are A, C, F, I. Hints: “the first jump has to land in the corner, from the right or from below”; “after that, which checker can jump into the new empty square?” Expected wrong report: “I left three”; ask for a replay. The cup check: start 8, so four left means four jumps.

Problem 8 (soldiers, the star one row above the line)

Two soldiers, in the star’s column, in the two rows below the line: one jump. One soldier can never jump. Insist that unused soldiers count and must come off the board; the answer is the smallest army that does it.

Problem 9 (the star two rows above the line)

Four soldiers, and up to mirror image there is only one army that works: three in a row in the row just below the line, ending in the star’s column, plus one directly below that end. The jumps: the lowest soldier jumps up over the one above it into the first row above the line; the far soldier of the three jumps sideways into the square just vacated, under the star’s column; that soldier jumps up over the one above the line onto the star.



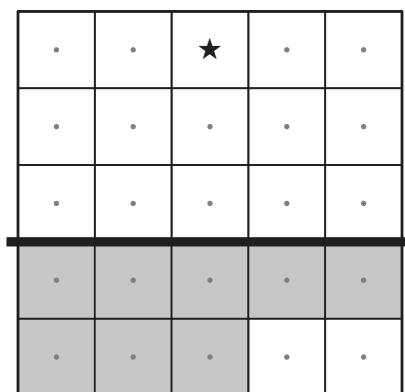
Why three cannot do it: the last jump is made by a soldier standing just below the line in the star’s column, over a soldier just above the line. Getting a soldier above the line already costs two soldiers (one jumps, one is removed), and it empties the square below the line that the jumper passed through, so a fresh soldier has to arrive there by a jump from the side, which costs two more. Hint to back into the shape: “which soldier makes the last jump, and where must the soldier he jumps over be standing?” Children usually find the L in a few minutes of trying. The prices so far: star 1 costs 2, star 2 costs 4. Do not reveal the next prices.

Problem 10 (reserve: the two-by-three boards)

Top board (hole in the middle of the top row): no jump at all; 0 in the box, five circles. Bottom board (hole in a corner): exactly one jump, along the top row into the corner, then stuck with four; 1 in the box, four circles. Why: an up-and-down jump needs three squares in a column, and this board has only two rows; the only lines of three are the two rows.

Problem 11 (reserve: the star three rows above the line)

Eight soldiers, seven jumps. On this board, which has two rows below the line, the only army that works is the block: five in the row just below the line, centred under the star, plus three in the row below them under the left three (or, in mirror image, under the right three).



Narrate it if needed; with columns called 1 to 5 from the left (the star is in column 3), rows $-1, 0$ below the line and 1, 2, 3 above: soldier (row -1 , col 2) jumps up over (0, 2) to (1, 2); soldier $(-1, 3)$ jumps up over (0, 3) to (1, 3); soldier (0, 5) jumps left over (0, 4) to (0, 3); soldier $(-1, 1)$ jumps up over (0, 1) to (1, 1); soldier (0, 3) jumps up over (1, 3) to (2, 3); soldier (1, 1) jumps right over (1, 2) to (1, 3); soldier (1, 3) jumps up over (2, 3) onto the star. The prices are now 2, 4, 8; a child will guess 16 for the next row. The true price of the fourth row is 20, and the fifth row can never be reached by any army, however large. Tell them so, as a fact from the grown-ups' table, if they ask.

Problem 12 (reserve: pennies on a strip, the line after square 6)

Top strip (star on square 7, just past the line): two pennies, on 5 and 6, and one jump; 2 goes in the box. The other strips (star on 8): no army ever gets a penny onto square 8; square 7 is as far as anyone gets, so square 7 is the one they colour. With all six pennies the play is forced: $5 > 6 > 7$, $3 > 4 > 5$, $1 > 2 > 3$, then pennies on 3, 5 and 7 and no jump left. Why, for a child: to land on 8 a penny must jump from 6 over a penny on 7; the penny on 7 got there by jumping from 5 over 6, which emptied 5 and 6; to refill 6 someone must jump from 4 over 5, so 5 must be refilled first, from 3 over 4, and so on: the army runs out. The grown-up version (a proof with numbers) is in Section 9.

If the K–1 table stalls or finishes early

Stalls come from the rule, not the mathematics: check the removed pieces and the “over a neighbour, into the empty square right behind” wording, and replay the launch strip. A child who is tired of strips can go

straight from Problem 5 to the soldiers (Problems 8–9), which are the most popular, and come back to the square boards. A fast table takes the reserves in the order 10, 11, 12. After that: “the star two rows up, but with the star in the corner column instead of the middle” (still four soldiers, the same L against the edge); “the strip of six with the hole on 2, but make the last checker end on the other square”; and the cup question on every board.

6. The grades 2–3 table (four third graders)

How the table runs. Children can read the problems, but read each new one aloud anyway and check the setup before they play. Work in pairs on the strips (each child records in their own packet), and let pairs compare tables before the “why” questions. The notation $3>2>1$ (the penny on 3 jumps over 2 onto 1) is introduced in Problem 2 and used from then on. A child who finishes a strip before the others tries to make the last penny end on the other square.

Timing (about 45 minutes). Problems 1–2: 5 minutes. Problems 3–4: 8. Problems 5–7: 14 (two pairs: one pair takes the six- and seven-strips, the other the eight-strip; they swap records at the end). Problems 8–11: 10. Problems 12–14: 8. The core ends with Problem 14 (the arrows of the star-2 army) at about 45 minutes, so it may be the share that interrupts it; that is fine. Everything from Problem 15 on is reserve, in the printed order: 15–17 (the army on the strip), 18–19 (the numbered strips, which are the proof of 15–17), 20, 21, 22–23 (the square boards), 24, 25, 26.

Problems 1–2 (strips of four, and the trees)

Hole 1: two left; hole 2: one, on square 3; hole 3: one, on 2; hole 4: two. The trees: hole 1 has a single game, $3>2>1$ and stuck (the second small strip stays empty); hole 2 has a single game, $4>3>2$, $1>2>3$. “How many games” is 1 and 1. The point of drawing them: a game with no choices is a proof of whatever it shows.

Problems 3–4 (strips of five, and the complete tree)

Every hole leaves two; none can be finished. The tree, which the children draw: hole 1: $3>2>1$, then $5>4>3$, stuck (pennies on 1 and 3). Hole 2: $4>3>2$, then $1>2>3$, stuck (3 and 5). Hole 3: $1>2>3$ then $4>3>2$, stuck (2 and 5); or $5>4>3$ then $2>3>4$, stuck (1 and 4). Four games in all (1, 1, 2), every one ending with two pennies. Say what they have done: they have not just failed to finish the strip, they have written down every possible game and none finishes it, which is a proof. Hint: “draw every possible first jump; then, for each one, every possible next jump.”

Problems 5–7 (strips of six, seven, eight)

The seven- and eight-strip tables have rows for holes 1–4 only; the other holes are their mirror images. Six: hole 1: 3 left ($3>2>1$, $5>4>3$; stuck on 1, 3, 6); hole 2: finished, last penny on 2 or 5 ($4>3>2$, $1>2>3$, $6>5>4$, then $3>4>5$ or $4>3>2$); holes 3 and 4: 2 left ($1>2>3$, $4>3>2$, $6>5>4$; stuck on 2 and 4); hole 5: finished, on 2 or 5; hole 6: 3 left. Seven: nothing works; holes 1, 4 (and 7) leave 3 at best, holes 2, 3 (and 5, 6) leave 2. The best lines: hole 1: $3>2>1$, $5>4>3$, $7>6>5$, stuck on 1, 3, 5 (the only game); hole 2:

4>3>2, 1>2>3, 6>5>4, 3>4>5, stuck on 5 and 7; hole 3: 5>4>3, 2>3>4, 7>6>5, 4>5>6, stuck on 1 and 6; hole 4: 2>3>4, 5>4>3, 7>6>5, stuck on 1, 3, 5. Eight: holes 2 and 5 finish with the last penny on 4 or 7 (from hole 2: 4>3>2, 1>2>3, 6>5>4, 3>4>5, 8>7>6, then 5>6>7 for 7 or 6>5>4 for 4); holes 4 and 7 finish with the last penny on 2 or 5 (from hole 4: 6>5>4, 3>4>5, 1>2>3, 8>7>6, 6>5>4, then 3>4>5 for 5 or 4>3>2 for 2); hole 3 (and 6) leaves 3: 1>2>3, 4>3>2, 6>5>4, 8>7>6, stuck on 2, 4, 6; hole 1 (and 8) leaves 4: 3>2>1, 5>4>3, 7>6>5, stuck on 1, 3, 5, 8, the only game. The three summary lines at the foot of Problem 7: lengths four, six and eight could be finished; the holes that worked were 2 (and 3) on four, 2 (and 5) on six, 2 and 4 (and 5, 7) on eight; five and seven never could. (For the adult: three works, nine does not, ten does; even lengths from four on have working holes, odd lengths from five on never.) Hint for the eight-strip: “finish the left half first, then the right.”

Problems 8–9 (three colours: count, jump, count)

Red squares are 1 and 4, blue 2 and 5, green 3 and 6. Hole 2 at the start: red 2, blue 1, green 2, total 5. Then, jump by jump: 4>3>2 gives red 1, blue 2, green 1 (total 4); 1>2>3 gives 0, 1, 2 (3); 6>5>4 gives 1, 0, 1 (2); the last jump, either one, gives 0, 1, 0 (1). Hole 5 (Problem 9, on its own strip) starts 2, 1, 2 as well and runs the same way in mirror image. What they should notice, and say: *every jump changes each of the three counts by exactly one* (two go down, one goes up), because the three squares of a jump always have three different colours. Ask: “if the three counts are all even now, what are they after one jump? And after the next?” (all odd, then all even, and so on: they are always all odd or all even together).

Problem 10 (the counts on the finished and unfinished strips)

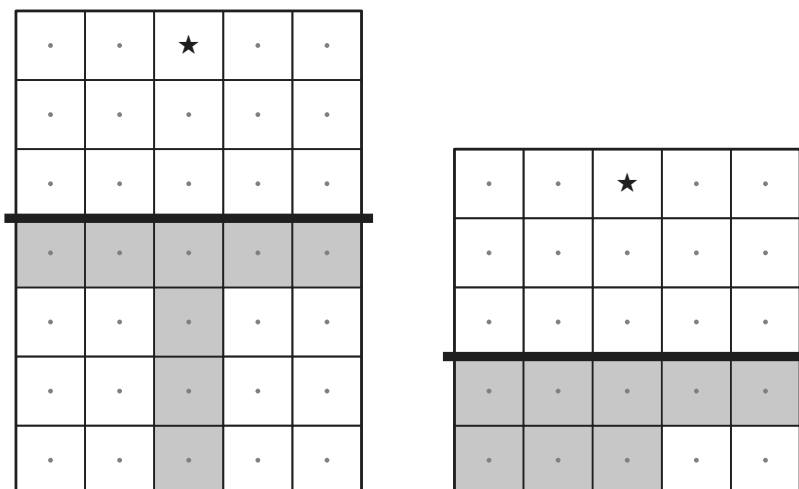
Four, hole 1: 1, 1, 1, no. Four, hole 2: 2, 0, 1, yes, last penny on green (square 3). Five, hole 3: 2, 2, 0, no. Five, hole 1: 1, 2, 1, no. Seven, hole 4: 2, 2, 2, no. Eight, hole 3: 3, 3, 1, no. Eight, hole 2: 3, 2, 2, yes, last penny on red (4 or 7). Eight, hole 4: 2, 3, 2, yes, blue (2 or 5). The rule to draw out, in the children’s words: *if the three counts are all odd or all even, the strip can never be finished*, because one penny alone has counts 1, 0, 0 (one odd, two even), and the counts stay all-odd or all-even forever. And when a strip can be finished, the last penny sits on the colour whose count is the odd one out at the start. Then point at five, hole 1: the counts 1, 2, 1 are *not* all alike, yet the strip cannot be finished (the tree of Problem 4 proved that). So the colours can prove “impossible”, but they can never promise “possible”.

Problem 11 (predictions)

All four strips are holes the children have not played (they are mirror images of rows in their tables, but do not say so until afterwards). Eight, hole 6: counts 3, 3, 1, cannot be finished; playing it leaves 3 at best (8>7>6, 5>6>7, 3>4>5, 1>2>3, stuck on 3, 5, 7). Eight, hole 5: counts 3, 2, 2; if it finishes, the last penny is on red (1, 4 or 7); playing it ends on 4 or 7 (3>4>5, 1>2>3, 6>5>4, 3>4>5, 8>7>6, then 5>6>7 or 6>5>4). Seven, hole 7: 2, 2, 2, cannot; 3 left at best (5>6>7, 3>4>5, 1>2>3). Eight, hole 8: 3, 2, 2; the colours allow red (1, 4, 7); playing it leaves 4 at best (6>7>8, 4>5>6, 2>3>4, stuck on 1, 4, 6, 8), so this prediction “maybe, on red” comes out “no”. Two predictions of impossibility came true, the one prediction of a colour came true, and the fourth strip shows the limit of the test.

Problems 12–14 (soldiers: stars one, two and three rows up)

Star 1: two soldiers under it. Star 2: four, the L (three in a row just below the line ending under the star, plus one below the end); its jumps in words: the bottom one jumps up over the one above it; the far one of the three jumps sideways into the vacated square; that one jumps up onto the star. Star 3: eight, and there are exactly two armies up to mirror image: the T (five in a row just below the line, centred under the star, and three more in the star's column going down) and the block (the same five, plus three in the next row down under the left three).



T, with columns 1–5 (star in column 3) and rows -3 to 0 below the line: $(-1,3)$ up over $(0,3)$ to $(1,3)$; $(0,1)$ right over $(0,2)$ to $(0,3)$; $(0,3)$ up over $(1,3)$ to $(2,3)$; $(0,5)$ left over $(0,4)$ to $(0,3)$; $(-3,3)$ up over $(-2,3)$ to $(-1,3)$; $(-1,3)$ up over $(0,3)$ to $(1,3)$; $(1,3)$ up over $(2,3)$ onto the star. Block: $(-1,2)$ up to $(1,2)$; $(-1,3)$ up to $(1,3)$; $(0,5)$ left to $(0,3)$; $(-1,1)$ up to $(1,1)$; $(0,3)$ up to $(2,3)$; $(1,1)$ right over $(1,2)$ to $(1,3)$; $(1,3)$ up onto the star. Problem 14 (the arrows for star 2): three arrows, numbered in the order above. The guess for star 4 will be 16. Tell them the truth only after they have guessed and tried a little: it is 20 (and that is a theorem, checked by computer), and a star five rows up can never be reached by any army of any size, which is the theorem the grown-ups' table is proving with weights. Why sixteen is hopeless is in Section 9.

Problems 15–17 (the army on a strip)

The farthest square any penny ever reaches is 7, the square just past the line, whatever army is placed on squares 1–6. The smallest army that gets past the line is two pennies, on 5 and 6. Square 8 is never reached. Problem 16, the army on 3, 4, 5, 6, has exactly four games: $4>3>2$ then $5>6>7$ (stuck on 2 and 7) or $6>5>4$ (stuck on 2 and 4); $5>6>7$ then $3>4>5$ (stuck on 5 and 7) or $4>3>2$ (stuck on 2 and 7). The farthest square in any of them is 7. Problem 17: with all six pennies there is only one possible first jump, $5>6>7$, then only $3>4>5$, then only $1>2>3$; the boxes get 1 and 0: no choice anywhere, and it ends stuck with pennies on 3, 5, 7. The argument for “never 8”: to land on 8 you need pennies on 6 and 7 at once; the penny on 7 came from 5 over 6, which emptied 5 and 6; to refill 6 you need 4 and 5 filled, so 5 must be refilled first by 3 over 4, and so on; every refill costs two pennies from farther left, and the army is finite. The full proof with numbers (Fibonacci labels) is Problems 18–19 and Section 9; at the share, the grades 4–5 table will show it.

Problems 18–19 (the numbered strips)

Four pennies: total $1 + 1 + 2 + 3 = 7$, star 8; the play is forced (the 2 jumps over the 3 onto the 5, then the 1 over the 1 onto the 2), the total stays 7 after each jump, and the game ends after two jumps, so the “after jump 3” column stays empty. Five pennies: total 12, star 13; forced again (the 3 over the 5 onto the 8, then the 1 over the 2 onto the 3), the total stays 12, two jumps, the last two columns stay empty. The star is always one more than the whole army, and a jump toward the star never changes the total (two neighbouring numbers add up to the next one), so no penny ever stands on the star.

Reserves

Problem 20 (how many ways). Exactly four winning games from hole 2 on the six-strip: $4 > 3 > 2$ is forced; then $1 > 2 > 3$ or $6 > 5 > 4$; then the other one is forced; then $3 > 4 > 5$ or $4 > 3 > 2$. **Problem 21 (blocks).** Fewest left: block of three, 2; four, 2; five, 3; six, 3. The block of three has exactly two games, $5 > 4 > 3$ (stuck on 3 and 6) and $5 > 6 > 7$ (stuck on 4 and 7). **Problem 22 (three-by-three).** Board A (centre hole): no jump at all. Board B (edge hole): one jump, the bottom-middle over the centre into the hole, and then 7 left. Board C (corner hole): fewest 4; a line, with squares A B C / D E F / G H I: $C > B > A$, $H > E > B$, $A > B > C$, $G > D > A$, leaving A, C, F, I. Three is impossible (every one of the fourteen complete games ends with four or five). The colouring gives nothing here (all three counts are different), so “we tried everything” is the only proof, and it is a real one. **Problem 23 (three-by-four with letters).** Hole A: can be finished, ten jumps, and the last penny can end on A or on D but nowhere else. One game ending on D: $C > B > A$, $J > F > B$, $A > B > C$, $D > C > B$, $H > G > F$, $E > F > G$, $L > K > J$, $I > J > K$, $K > G > C$, $B > C > D$; replace the last jump by $C > B > A$ to end on A. Hole B: can be finished, and the last penny can only end on B: $D > C > B$, $A > B > C$, $I > E > A$, $J > F > B$, $B > C > D$, $H > G > F$, $L > K > J$, $J > F > B$, $A > B > C$, $D > C > B$. Hole E (end of the middle row): one jump only ($G > F > E$), then 10 left. Hole F: at most two jumps ($H > G > F$, then $E > F > G$), 9 left. Moving the hole one square turns a board with 852 winning games into one with none. Hint for A: “clear the top row first, using the pennies below it.” **Problem 24 (three-by-five).** Only the hole on C (the middle of a long side; H, the very centre, does not work) can be finished, in thirteen jumps, and the last penny returns to C: $A > B > C$, $D > C > B$, $K > F > A$, $A > B > C$, $L > G > B$, $B > C > D$, $E > D > C$, $I > H > G$, $N > M > L$, $L > G > B$, $B > C > D$, $O > J > E$, $E > D > C$. Holes A and B leave 2, and a child will want to see it done: hole A: $C > B > A$, $E > D > C$, $L > G > B$, $B > C > D$, $I > H > G$, $F > G > H$, $N > M > L$, $K > L > M$, $M > H > C$, $D > C > B$, $A > B > C$, $O > J > E$ leaves C and E; hole B: $D > C > B$, $A > B > C$, $K > F > A$, $L > G > B$, $B > C > D$, $E > D > C$, $I > H > G$, $N > M > L$, $L > G > B$, $B > C > D$, $O > J > E$, $E > D > C$ leaves A and C. Hole H leaves 12. **Problem 25 (the cross).** Two left at best from any hole, and never one. Hole at the end of the right arm: $\text{centre} > \text{right1} > \text{right2}$, $\text{down2} > \text{down1} > \text{centre}$, $\text{left1} > \text{centre} > \text{right1}$, $\text{up2} > \text{up1} > \text{centre}$, $\text{right1} > \text{centre} > \text{left1}$, $\text{left2} > \text{left1} > \text{centre}$ leaves two (the centre and right2). Hole in the centre: $\text{down2} > \text{down1} > \text{centre}$, $\text{up1} > \text{centre} > \text{down1}$, $\text{left2} > \text{left1} > \text{centre}$, $\text{down1} > \text{centre} > \text{up1}$, $\text{right2} > \text{right1} > \text{centre}$, $\text{up1} > \text{centre} > \text{down1}$ leaves two (down1 and up2). Why never one: from an arm-end hole there are only nine complete games (four leave 2, four leave 3, one leaves 6), and a pair can list all nine; from the centre there are thirty-two (sixteen leave 2, eight leave 3, eight leave 4). For the mathematician: the diagonal colouring $(r + c) \bmod 3$ certifies the centre hole (classes 1, 4, 4 with the centre removed become 0, 4, 4, all even) but is silent for the arm-end hole (1, 4, 3). **Problem 26 (rings).** Fewest left: ring of 5: 2; ring of 6: 1; ring of 7: 2; ring of 8: 1. With the cells numbered 1 to n clockwise from the hole: ring 5: $3 > 2 > 1$, $5 > 1 > 2$, stuck on 2 and 4 (there are only four complete games, all stuck with two). Ring 6: $3 > 2 > 1$, $5 > 4 > 3$, $6 > 1 > 2$, $2 > 3 > 4$, finished on 4 (eight winning games). Ring 7: $3 > 2 > 1$, $5 > 4 > 3$, $7 > 1 > 2$, $2 > 3 > 4$, stuck on 4 and 6 (twelve complete games, none finishes). Ring 8: $3 > 2 > 1$, $5 > 4 > 3$, $7 > 6 > 5$, $8 > 1 > 2$, $2 > 3 > 4$, $4 > 5 > 6$, finished on 6 (forty winning games). Even rings can be finished, odd rings cannot (this continues: 9 and 11 cannot, 10 and 12 can). Why the colours do not settle it: the strip’s three-colouring closes up around a ring only when n is a multiple of 3, and on the ring

of 9 it is silent (counts 2, 3, 3), so for 5 and 7 the proof is the exhaustive tree.

If the grades 2–3 table stalls or finishes early

A pair that cannot finish the eight-strip (Problem 7) should be shown the first three jumps for hole 2 and left to find the rest. If the colouring (Problems 8–11) is not landing, go back to the four-strip with the hole at the end: 1, 1, 1 at the start, and ask what one jump does. A table that finds the soldiers more fun than the colours should go to Problems 12–14 after Problem 7 and return to the colours with the data of the eight-strip. A fast table takes the reserves in the printed order: Problems 15–17, then 18–19 (the numbered strips are the proof of 15–17), then 20, 21, 22–23, 24, 25, 26. After all that: “on the strip of eight, from hole 2, how many different winning games are there?” (ten; the tree is written out under Problem 24 of Section 7, and a pair can do it), and “a strip of ten, which holes work?” (2, 5, 6, 9).

7. The grades 4–5 table (two fourth graders, one fifth grader)

How the table runs. These children can run the experiments themselves; your job is to insist on complete records, to ask “how do you know for sure?” at the right moments (marked below), and to move them on before the strips eat the hour. Split the data collection: one child takes the four- and five-strips, one the six, one the seven and eight, then pool. A child waiting for the pooled data writes the ending squares of the mirror holes (hole 3 of four, 4 and 5 of five, 4 to 6 of six, 5 to 7 of seven, 5 to 8 of eight) under the tables. The core is Problems 1–10, and it fills the forty minutes; the numbered strip and its explanation (Problems 8–10) are the heart of the session, so start Problem 8 by 0:35 at the latest, even if the soldiers are unfinished.

Timing (about 40 minutes). Problems 1–2: 10 minutes (pool at 0:18). Problems 3–5: 9. Problems 6–7: 8. Problems 8, 9, 10: 13. The core ends with Problem 10 (the written explanation). Everything else is reserve in the printed order: 11 (the star-2 weights), 12 (the whole-number weights, which refute the guess of 16), 13 (the star-3 weights), 14–17 (what the colouring can and cannot do; reversal), 18–19, 20–24. The star-2 weights of Problem 11 can be shown orally at the share if the table never reaches them.

Problems 1–2 (strips of four to eight)

Fewest left, and ending squares when one is left. Four: hole 1: 2; hole 2: 1, on 3. Five: holes 1, 2, 3: 2, 2, 2. Six: hole 1: 3; hole 2: 1, on 2 or 5; hole 3: 2. Seven: holes 1, 2, 3, 4: 3, 2, 2, 3. Eight: hole 1: 4; hole 2: 1, on 4 or 7; hole 3: 3; hole 4: 1, on 2 or 5. Mirror holes are the same with the squares reflected ($k \mapsto n + 1 - k$): on eight, hole 5 ends on 4 or 7 and hole 7 ends on 2 or 5. The summary lines: lengths four, six and eight can be finished, from the holes next to the ends (2, and its mirror), and on eight also from 4 and 5; five and seven never. Games: six, hole 2: $4 > 3 > 2$, $1 > 2 > 3$, $6 > 5 > 4$, then $3 > 4 > 5$ (ends on 5) or $4 > 3 > 2$ (ends on 2). Eight, hole 2: $4 > 3 > 2$, $1 > 2 > 3$, $6 > 5 > 4$, $3 > 4 > 5$, $8 > 7 > 6$, and then either $5 > 6 > 7$ (ends on 7) or $6 > 5 > 4$ (ends on 4). Eight, hole 4: $6 > 5 > 4$, $3 > 4 > 5$, $1 > 2 > 3$, $8 > 7 > 6$, $6 > 5 > 4$, $3 > 4 > 5$ (ends on 5), or end with $4 > 3 > 2$ for 2. Best lines for the holes that fail, in case a child doubts the fewest: six, hole 1: $3 > 2 > 1$, $5 > 4 > 3$ leaves 1, 3, 6; six, hole 3: $1 > 2 > 3$, $4 > 3 > 2$, $6 > 5 > 4$ leaves 2, 4; seven, hole 1: $3 > 2 > 1$, $5 > 4 > 3$, $7 > 6 > 5$ leaves 1, 3, 5; seven, hole 2: $4 > 3 > 2$, $1 > 2 > 3$, $6 > 5 > 4$, $3 > 4 > 5$ leaves 5, 7; seven, hole 3: $5 > 4 > 3$, $2 > 3 > 4$, $7 > 6 > 5$, $4 > 5 > 6$ leaves 1, 6; seven, hole 4: $2 > 3 > 4$, $5 > 4 > 3$, $7 > 6 > 5$ leaves 1, 3, 5; eight, hole 1: $3 > 2 > 1$, $5 > 4 > 3$, $7 > 6 > 5$ leaves

1, 3, 5, 8 (the only game from that hole); eight, hole 3: $1>2>3$, $4>3>2$, $6>5>4$, $8>7>6$ leaves 2, 4, 6. Make sure they see both endings for hole 2 and hole 4; Problem 15 needs them.

Problems 3–5 (the colouring, and the theorem)

Red = squares 1, 4, 7; blue = 2, 5, 8; green = 3, 6. Six, hole 2: start 2, 1, 2 (total 5); after $4>3>2$: 1, 2, 1; after $1>2>3$: 0, 1, 2; after $6>5>4$: 1, 0, 1; after the last jump: 0, 1, 0. Problem 4 table (red, blue, green at the start; finished; ending colour): four/1: 1,1,1, no. Four/2: 2,0,1, yes, green. Five/1: 1,2,1, no. Five/3: 2,2,0, no. Six/1: 1,2,2, no. Six/2: 2,1,2, yes, blue. Six/3: 2,2,1, no. Seven/1: 2,2,2, no. Seven/2: 3,1,2, no. Eight/1: 2,3,2, no. Eight/2: 3,2,2, yes, red. Eight/3: 3,3,1, no. Eight/4: 2,3,2, yes, blue.

Problem 5: (a) every jump changes each count by exactly one: the three squares of any jump are three consecutive squares, so they carry all three colours; two of them lose a penny and one gains. (b) A single penny on blue: 0, 1, 0. (c) All odd or all even: four/1, five/3, seven/1, eight/3; none was finished. (d) The ending colour is always the odd-one-out colour at the start. Now ask for the proof “in one breath”, and help them assemble it: *each jump flips the parity of all three counts together, so the three counts stay all-odd or all-even, or else keep the same one colour odd and two even; a single penny is one odd and two even; so from all-odd or all-even you can never reach one penny, and otherwise the last penny must be on the colour that started with the odd-one-out parity.* That is a theorem, and they proved it. If the proof does not come, ask the three questions in order: “what colours are the three squares of a jump?”, “so what happens to each count?”, “what are the counts when one penny is left?”

Problems 6–7 (soldiers)

Star 1: 2 soldiers (e4, e3): $e3>e4>e5$. Star 2: 4, the L (c4, d4, e4, e3) or its mirror: $e3>e4>e5$, $c4>d4>e4$, $e4>e5>e6$. Star 3: 8, the T (c4, d4, e4, f4, g4, e3, e2, e1): $e3>e4>e5$, $c4>d4>e4$, $e4>e5>e6$, $g4>f4>e4$, $e1>e2>e3$, $e3>e4>e5$, $e5>e6>e7$; or the block (c4, d4, e4, f4, g4, c3, d3, e3): $d3>d4>d5$, $e3>e4>e5$, $g4>f4>e4$, $c3>c4>c5$, $e4>e5>e6$, $c5>d5>e5$, $e5>e6>e7$. These are all the smallest armies up to mirror image. Hint for star 2: “which soldier makes the last jump, and where must the soldier it jumps over be standing?” Hints for star 3, the hardest search of the session, in order: “the last jump is vertical in the star’s column; what two squares must be full just before it?” (e5 and e6); “the soldier on e6 got there by a jump from e4 over e5; that emptied e4 and e5, so e5 must be refilled, from the side or from below”; “five in a row just below the line, plus three more: try a column, then try a second row.” The guess for star 4 is 16; the truth is 20 (Problem 20 has an army and its nineteen jumps), and sixteen soldiers cannot even come close: the weights of Problem 12 show it. Do not give away 20 until Problem 12.

Problems 8–10 (the numbered strip)

The labels are 1, 1, 2, 3, 5, 8 left of the line, then 13, then the star 21. Problem 8: (a) pennies on 5 and 8: total 13; after the jump onto 13, total 13: unchanged. (b) After 8 jumps over 5 onto 3: total 3, down by 10. (c) All six: total 20; the star is 21. The play is forced: 5 over 8 onto 13 (total 20), then 2 over 3 onto 5 (20), then 1 over 1 onto 2 (20), then stuck with pennies on 2, 5, 13. The largest total ever is 20. (d) No army on the six squares reaches the star; the farthest square is always the 13, just past the line (the two-penny army on 5 and 8 gets there in one jump), and the mini strips record which armies they tried.

Problem 9: four pennies: 7 against 8; five: 12 against 13; six: 20 against 21; seven: 33 against 34; eight: 54 against 55; the difference is 1 every time. The farthest square reached is always the one just past the

line: 5, 8, 13, 21, 34 (its label is the sum of the army's last two labels, and the jump of the second-last penny over the last one lands there); the star, one square further, is never reached.

Problem 10, the argument they should write: a jump toward the star replaces two neighbouring labels by the next label, which is their sum, so the total does not change; a jump away from the star replaces two neighbours by the label before them, which is smaller, so the total goes down; therefore the total never goes up. It starts at the army's total, which is one less than the star on every strip. A penny on the star would by itself contribute the star's number. So no penny ever stands on the star. This is a complete proof of "one square past the line and never two", by addition alone. It is also exactly the skeleton of the grown-up theorem in Problems 11–13 and Section 9.

Problems 11–13 (the golden weights and the whole-number weights)

Problem 11 (star 2): (a) $0.618 + 0.382 = 1.000$, $0.382 + 0.236 = 0.618$, $0.236 + 0.146 = 0.382$, $0.146 + 0.090 = 0.236$: two neighbouring weights along a line toward the star add up to the next one, exactly as on the numbered strip. (b) The L weighs $0.382 + 0.236 + 0.146 + 0.236 = 1.000$, the star's weight; after each of the three jumps the total is still 1.000. (c) The three heaviest squares below the line weigh $0.382 + 0.236 + 0.236 = 0.854 < 1$, so three soldiers can never reach star 2: the total never goes up, and a soldier on the star is worth 1.000 alone.

Problem 12 (star 4, whole numbers): the weights are Fibonacci numbers, 144 on the star and F two steps down the sequence for each step away, so that along any line toward the star two neighbours add up to the next ($3 + 5 = 8$, $8 + 13 = 21$, $34 + 55 = 89$, in every direction). (b) The twenty shaded squares weigh $79 + 34 + 23 + 8 = 144$ (row by row from the line down), exactly the star. (c) The sixteen heaviest below the line: $21 + 3(13) + 5(8) + 7(5) = 135 < 144$, so the line reads "sixteen soldiers cannot reach star 4, because the sixteen heaviest squares together weigh only 135 and a soldier on the star weighs 144 by itself"; the guess of 16 is refuted by arithmetic. (d) The nineteen heaviest: $135 + 3(3) = 144$, exactly the star, so the weight rule alone does not settle nineteen; a computer search over every nineteen-soldier army that weighs enough shows that none works, so the answer is 20. The decimal weights of Problems 11 and 13 are the same numbers scaled: 144 times 0.618^d is close to the whole number at distance d (e.g. $144 \times 0.146 = 21.0$).

Problem 13 (star 3): (a) the T weighs $0.236 + 2(0.146) + 2(0.090) + 0.146 + 0.090 + 0.056 = 1.000$. (b) The block weighs $0.236 + 2(0.146) + 2(0.090) + 0.056 + 0.090 + 0.146 = 1.000$. (c) The seven heaviest weigh $0.236 + 3(0.146) + 3(0.090) = 0.944 < 1$, so seven soldiers cannot reach star 3. (d) Playing the T, the total stays 1.000 after every jump: every jump goes straight toward the star.

Reserves

Problem 14 (what the certificate does and does not do). Eight, hole 2: red is the odd one out (3 against 2, 2), so the colouring allows the last penny on 1, 4 or 7; the data show 4 and 7 only, never 1. The colouring is a true statement about every game, but it allows more than the game delivers. Ten, hole 1: red 3 (1, 4, 7, 10 minus the hole), blue 3, green 3; all odd; impossible, and playing it confirms it: 5 left at best ($3 > 2 > 1$, $5 > 4 > 3$, $7 > 6 > 5$, $9 > 8 > 7$, the only game, stuck on 1, 3, 5, 7, 10). Nine, hole 5: red 3, blue 2, green 3; the colouring says "if it finishes, on blue"; the truth is that no hole on a strip of nine can be finished (2 left at best; from hole 5: $3 > 4 > 5$, $1 > 2 > 3$, $6 > 5 > 4$, $3 > 4 > 5$, $8 > 7 > 6$, $5 > 6 > 7$ leaves 7 and 9), and the colouring cannot see it. The moral, which the fifth grader should be able to state: a colouring can certify "impossible"; it can never certify "possible"; and when it is silent, a complete tree of plays is the only proof.

Problem 15 (pairs). On eight: hole 2 ends on 4 or 7; hole 4 on 2 or 5; hole 5 on 4 or 7; hole 7 on 2 or 5. The pairs: (2,4), (2,7), (4,5), (5,7); every relation “hole a can end on b ” is matched by “hole b can end on a ”, and on six, (2,5) likewise.

Problems 16–17 (reversal). Problem 16: a game from hole 2 ending on 4 is $4>3>2$, $1>2>3$, $6>5>4$, $3>4>5$, $8>7>6$, $6>5>4$; the same six jumps last-first, $6>5>4$, $8>7>6$, $3>4>5$, $6>5>4$, $1>2>3$, $4>3>2$, are all legal from the hole on 4 and end on 2. On six: $4>3>2$, $1>2>3$, $6>5>4$, $3>4>5$ ends on 5; reversed from hole 5, $3>4>5$, $6>5>4$, $1>2>3$, $4>3>2$ ends on 2. Problem 17: from hole A ending on D, $C>B>A$, $J>F>B$, $A>B>C$, $D>C>B$, $H>G>F$, $E>F>G$, $L>K>J$, $I>J>K$, $K>G>C$, $B>C>D$; the same ten jumps in reverse order, from hole D, end on A.

Why it works (ask them, then give it): write down the positions of the winning game and swap pegs and holes in every one. The final position (one penny on b) becomes “every square full except b ”, the start becomes “one penny on a ”. A jump $x>y>z$ turns a position with pennies on x, y and a hole on z into one with holes on x, y and a penny on z ; after swapping, that same jump $x>y>z$ leads from the swapped later position back to the swapped earlier one. So the swapped positions, read backwards, are a legal game from hole b to a single penny on a . This is the reversal principle of peg solitaire (Section 9).

Problems 18–19 (sixteen soldiers; the star-2 jumps). Problem 18: with sixteen the best they will do is star 3 (eight soldiers already do that); “highest star reached” is the star number, 3 at best, and after Problem 12 the honest answer for star 4 is “19 or 20”. Problem 19: $e3>e4>e5$, $c4>d4>e4$, $e4>e5>e6$ (or the mirror image with $g4$, $f4$). **Problem 20 (the twenty-soldier army).** The list is correct and ends with $e6>e7>e8$; a soldier stands on $e8$ after nineteen jumps. After jump 16 the four soldiers left are $e5$, $e6$, $f6$, $g6$; then $e5$ jumps to $e7$ (emptying $e6$), $g6$ jumps over $f6$ to refill $e6$, and $e6$ jumps over $e7$ onto the star. **Problem 21.** Every jump in the list moves its soldier two squares closer to the star in the taxicab sense (six of them are horizontal, which the children will call sideways, but each still gets two squares closer); a jump of that kind keeps the total weight, every other jump lowers it. If the army weighs exactly the star’s weight and a jump ever lowered the total, the total would stay below the star’s weight forever, and a soldier on the star is worth the star’s weight by itself; so every jump must bring the jumper two steps closer to the star. **Problem 22 (four-by-four).** From hole B the board can be finished in fourteen jumps and the last penny ends on C or on O: $D>C>B$, $A>B>C$, $I>E>A$, $G>F>E$, $A>E>I$, $L>H>D$, $D>C>B$, $M>I>E$, $N>J>F$, $B>F>J$, $K>J>I$, $E>I>M$, $P>O>N$, $M>N>O$. Hole A with pattern 1: the squares marked 2 are the four corners plus G and J, six of them; with A empty the counts are 5, 5, 5, all odd, impossible. Hole F with pattern 2: the squares marked 0 are the two diagonals’ six squares (A, F, K, P, D, M); counts 5, 5, 5, impossible. Hole F with pattern 1: 5, 4, 6, no conclusion. Why a jump changes each count by one: a horizontal jump uses three squares with the same row and columns $c, c+1, c+2$, a vertical one the same column and rows $r, r+1, r+2$; either way $r+c$ takes three consecutive values, one of each class. Pattern 2 works the same way with $r-c$. The eight edge squares that are not corners are exactly the holes that can be finished; the four corners and the four centre squares cannot, and the two patterns together certify all eight impossibilities. Two-left lines, since a child will ask: hole A: $C>B>A$, $J>F>B$, $A>B>C$, $D>C>B$, $H>G>F$, $B>F>J$, $I>E>A$, $K>J>I$, $M>I>E$, $A>E>I$, $O>N>M$, $M>I>E$, $P>L>H$ leaves E and H. Hole F: $H>G>F$, $E>F>G$, $M>I>E$, $A>E>I$, $C>B>A$, $K>G>C$, $D>C>B$, $A>B>C$, $I>J>K$, $O>K>G$, $C>G>K$, $L>K>J$, $N>J>F$ leaves F and P. **Problem 23 (rings).** Fewest left: ring of 5: 2; 6: 1; 7: 2; 8: 1. With the cells numbered 1 to n clockwise from the hole: ring 5: $3>2>1$, $5>1>2$, stuck on 2 and 4. Ring 6: $3>2>1$, $5>4>3$, $6>1>2$, $2>3>4$. Ring 7: $3>2>1$, $5>4>3$, $7>1>2$, $2>3>4$, stuck on 4 and 6. Ring 8: $3>2>1$, $5>4>3$, $7>6>5$, $8>1>2$, $2>3>4$, $4>5>6$. Even rings can be finished, odd rings cannot (for the adult: this continues, 9 and 11 cannot, 10 and 12 can). Why the colours do not settle the odd rings: the strip’s three-colouring closes up around a ring only when n is a multiple of 3, and on the ring of 9 it is silent (counts 2, 3, 3), so for 5 and 7 the proof is the exhaustive tree: 4 complete games on the ring of 5, 12 on the ring of 7, none finishes. **Problem 24 (counting).** Four winning games on six from hole 2 (the tree of Section 6, Problem 20: ten positions in all, which is why twelve blank strips are printed). On eight from hole 2 there are fifteen complete games, ten winning and five stuck

with pennies on 2 and 6. By first three jumps: $4>3>2$, $1>2>3$, $6>5>4$: four winning games (then $3>4>5$ and $8>7>6$ in either order, then $5>6>7$ ending on 7 or $6>5>4$ ending on 4); $4>3>2$, $6>5>4$, $1>2>3$: four more, the same way; $4>3>2$, $6>5>4$, $8>7>6$: two ($1>2>3$, $3>4>5$, then $5>6>7$ or $6>5>4$). The five stuck games are the ones that play $4>3>2$ a second time before $3>4>5$. (And 28 winning games on ten from hole 2.)

If the grades 4–5 table stalls or finishes early

If the data collection is slow, stop it after the six-strip and give them the seven- and eight-strip results from this guide; the colouring argument (Problem 5) matters more than the data. If the proof in Problem 5 does not come, use the three questions listed there. If the soldiers absorb the table, set a timer: the numbered strip (Problem 8) must start by 0:35. A fast table takes the reserves in the printed order: 11, 12, 13, then 14–17, 18–19, 20 (execute the twenty), 21, 22, 23, 24. After that, two questions with no printed page: “why does the whole infinite half-plane weigh exactly 1 for a star five rows up?” (the row sums are 0.382, 0.236, 0.146, $\dots$, each two adding to the previous, total 1; Section 9) and “on the numbered strip, what if the army may also retreat to the left of square 1?” (label the extra squares to the left with the Fibonacci sequence continued backwards, 1, 0, 1, -1 , $\dots$; the sum rule still holds, but negative labels break the argument, so give the army more room only by using larger labels: Section 9).

8. The share (eight minutes)

Each table shows one thing, with pieces, in two minutes; the organizer closes.

- **K–1** shows the two four-strips: the hole at the end gets stuck with two checkers that are not neighbours; the hole next to the end finishes. Then a child sets up the L and gets a soldier two rows across the line, and says the prices: 2 soldiers for one row, 4 for two.
- **Grades 2–3** shows the colour rule on the four-strip with the hole at the end: red 1, blue 1, green 1; every jump changes every count by one; a single penny is 1, 0, 0; so never. Then the tree for the five-strip with the middle hole: two games, both stuck with two.
- **Grades 4–5** shows the numbered strip: the army adds to 20, the star is 21, and a jump toward the star keeps the total (5 over 8 onto 13). So the star is never reached. Then, for the whole room: 8 soldiers reach the third row (show the T), 20 reach the fourth, and the fifth row can never be reached.
- **The organizer closes** in three sentences: the weights on the big board are the same trick in two dimensions; if the star is five rows up, the whole infinite army below the line weighs exactly as much as the star and nothing more; so any real army, which is a part of it, weighs less, and can never get there. This is Conway’s theorem about soldiers, proved with a golden-ratio ruler in 1961, and it is the same idea a kindergartner used today when counting the cup.

9. The mathematics behind the session (for the two mathematicians)

The game. A position is a set of occupied cells; a jump is the local replacement $XX \rightarrow \dots X$ along a line. The reachability question, “can the position S be reduced to a single peg (on a given cell)?”, is a path question in a finite directed graph of positions; on general boards it is NP-complete (Uehara–Iwata 1990,

for a generalised Hi-Q), which is why certificates of impossibility matter: a winning sequence certifies “yes”, but “no” needs either exhaustion or an invariant.

Conservation. Each jump removes one peg: pegs at the start = jumps + pegs left. This is the K–1 invariant; the two below are its refinements.

Parity classes. Colour a strip 1, 2, 3, 1, 2, 3, . . . (cell k gets $k \bmod 3$). A jump occupies three consecutive cells, one of each colour, and changes each colour count by ± 1 , so the vector of parities (p_1, p_2, p_3) is flipped in all three coordinates by every jump; the class $\{(p_1, p_2, p_3), (\bar{p}_1, \bar{p}_2, \bar{p}_3)\}$ is invariant. A single peg has parity class $\{(1, 0, 0), (0, 1, 1)\}$ or a permutation, so from $(0, 0, 0)/(1, 1, 1)$ one peg is unreachable, and otherwise the last peg’s colour is determined. On the square grid the same works with the two diagonal colourings $(r + c) \bmod 3$ and $(r - c) \bmod 3$ (a horizontal or vertical jump runs through three consecutive values of both), giving de Bruijn’s and Conway’s sixteen position classes; column stripes fail because a vertical jump stays in one column class. On the 4×4 board the two colourings certify exactly the eight impossible holes; on 3×3 , 3×4 , 3×5 every hole passes both tests and the impossibilities are geometric (exhaustion). The strip data in Section 10 show how often the colouring is silent.

Pagoda (potential) functions. Give each cell a real weight w such that no jump increases the total weight of the occupied cells: $w(z) \leq w(x) + w(y)$ for every jump $x > y > z$. Then if a target position weighs more than the start, it is unreachable. This is a monovariant argument (Engel, *Problem-Solving Strategies*, ch. 1); in the linear-programming relaxation of reachability (the target minus the start must be a non-negative combination of jump vectors), a pagoda function is exactly a dual certificate, which is the honest answer to “where does σ come from”.

Conway’s soldiers. Let $\sigma = (\sqrt{5} - 1)/2 \approx 0.618$, the positive root of $\sigma^2 + \sigma = 1$, so $\sigma^{d+1} + \sigma^{d+2} = \sigma^d$. Weight a cell at taxicab distance d from the target by σ^d . Along a line the distances of three consecutive cells are $d, d + 1, d + 2$ (toward or away) or $d + 1, d, d + 1$ (crossing the target’s row or column). A jump toward the target replaces $\sigma^{d+2} + \sigma^{d+1}$ by σ^d : total unchanged; away: replaces $\sigma^d + \sigma^{d+1}$ by σ^{d+2} : a loss; crossing: replaces $\sigma^{d+1} + \sigma^d$ by σ^{d+1} : a loss. So the total never increases. For a target five rows above the line, the whole half-plane below the line weighs

$$\begin{aligned} \sum_{r \geq 0} \sum_{c \in \mathbb{Z}} \sigma^{5+r+|c|} &= \sigma^5 \cdot \frac{1}{1 - \sigma} \cdot \left(1 + \frac{2\sigma}{1 - \sigma}\right) = \sigma^3 (1 + 2/\sigma) = \sigma^3 + 2\sigma^2 \\ &= \sigma^2(\sigma + 2) = (1 - \sigma)(2 + \sigma) = 2 - \sigma - \sigma^2 = 1, \end{aligned}$$

using $1 - \sigma = \sigma^2$. The row sums are $\sigma^2, \sigma^3, \sigma^4, \dots = 0.382, 0.236, 0.146, \dots$ (each two adding to the previous), which is the version for the fast fifth grader. A finite army is a proper subset and weighs strictly less than 1, and a soldier on the target weighs exactly 1, so the fifth row is unreachable (Conway, 1961; Berlekamp–Conway–Guy, *Winning Ways*, vol. 4, ch. 23; Honsberger, *Mathematical Gems II*, has the shortest full exposition). Rows 1–4 are reachable, with minimal armies of 2, 4, 8, 20. The weights give the lower bounds directly: the m heaviest cells below the line weigh, for target row 2: 0.382, 0.618, 0.854, 1.090 for $m = 1..4$; for row 3: 0.944 for $m = 7$ and 1.034 for $m = 8$; for row 4: 0.936 for $m = 16$, 0.979 for $m = 18$, exactly 1 for $m = 19$, 1.021 for $m = 20$. So 3, 7, 18 soldiers are impossible by weight alone; 19 weighs exactly enough, which forces every jump to be straight toward the target and the army to be the sixteen cells at distance ≤ 7 plus three at distance 8, and a finite search over those 84 armies (209,848 positions) finds no solution; hence 20 (Berlekamp–Conway–Guy; Bell, Hirschberg and Guerrero-García, *INTEGERS* 7 (2007), G07, for minimal armies on this and other lattices). The 2-4-8 pattern breaks at the fourth case, to 20, and then to never. Every minimal army weighs exactly 1 (the L, the T and the block, the twenty), so none of them ever makes a wasteful jump; the number of jumps is always soldiers minus one: 1, 3, 7, 19.

The one-dimensional version and Fibonacci. On a strip the same argument works with integers: label the cells $F_1, F_2, \dots$ toward the star, so two neighbours add to the next, and $F_1 + \dots + F_k = F_{k+2} - 1$

(induction: add F_{k+1} to both sides). An army on the first k cells totals one less than the star at cell $k + 2$, while cell $k + 1$ is reachable ($F_{k-1} + F_k = F_{k+1}$ in one jump). This is Aigner's "Moving into the desert with Fibonacci" (*Math. Magazine* 70 (1997) 11–21), and it is why the 4–5 pages use whole numbers on the row-4 board: $w(d) = F_{N-d}$ satisfies $w(d+1) + w(d+2) = w(d)$ exactly, so it is a pagoda function on any board where all cells have $d \leq N - 1$ (positive weights); $N = 12$ gives 144 on the star and 1 in the farthest corner of the printed board, and the twenty-army weighs exactly 144 while the sixteen heaviest cells weigh 135. Since $F_{N-d}/F_N \rightarrow \sigma^d$, the integer weights are the golden weights rounded to a common scale, and the student pages' decimal sums for rows 2 and 3 are exact to three places because they are short; for row 4 the decimals would drift (the twenty-army sums to 0.997 with three-place weights), which is the practical reason for the integers. The reserve question "what if the army may retreat left of cell 1?": with $F_0 = 0, F_{-1} = 1, F_{-2} = -1$ the sum rule persists but the weights go negative, so instead label the star F_N with $N \geq 3k$; every cell a peg could ever reach then has a positive label, and the army still totals at most $F_N - F_{N-k} < F_N$. Aigner shows the one-dimensional army can never advance two cells past the line however much room it has; our search confirms it for every army on up to twelve cells with six cells of retreat.

Reversal. Complementing a position (pegs $\leftrightarrow$ holes) and reversing time turns a jump into a jump: if $P \xrightarrow{x>y>z} P'$ then $\overline{P'} \xrightarrow{x>y>z} \overline{P}$. Hence "hole a can end on b " is equivalent to "hole b can end on a ", which the children see in the strip data (2/4, 2/7, 4/5, 5/7 on the eight-strip; A/D on the 3×4 ; C/C on the 3×5). This is the classical reversal principle (Beasley, *The Ins and Outs of Peg Solitaire*).

Exhaustion and counting. On tiny boards the tree of all plays is a proof: the five-strip has four games, the six-strip from hole 2 has four winning games, the eight-strip from hole 2 has ten, ten from hole 2 has 28; the 3×4 from a corner has 852, from an edge-middle 488; the 3×5 from the middle of a long side 17,424; the 4×4 from an edge-middle 210,422. Rings: a ring of n cells with one hole can be finished for $n = 3, 4, 6, 8, 10, 12$ and not for 5, 7, 9, 11 (winning sequences 4, 8, 40, 112, 336 for $n = 4, 6, 8, 10, 12$).

Where it leads. Beeler and Hoilman, "Peg solitaire on graphs", *Discrete Math.* 311 (2011): which graphs (paths, cycles, stars, complete graphs, products) can be solved from a single hole; the strip and ring data above are its smallest cases. Csákány and Juhász, "The solitaire army reinspected", *Math. Magazine* 73 (2000), and Eriksson and Lindström, "Twin jumping checkers in $\mathbb{Z}^d$ ", *European J. Combin.* 16 (1995): higher dimensions and diagonal moves. De Bruijn, "A solitaire game and its relation to a finite field", *J. Recreational Math.* 5 (1972): the sixteen position classes as a vector space over $\mathbb{F}_4$. The monovariant style reappears as potential functions in amortised analysis of algorithms and as Lyapunov functions in dynamics. Semi-open questions the two of you may enjoy: the exact minimum number of pegs left by a block of n on a long strip (the data say $\lceil n/2 \rceil$ for $n \leq 12$; we know no proof), minimal armies when diagonal jumps are allowed, and solitaire on other lattices.

10. Reference tables

Strips with one hole. Fewest pegs that can be left; cells where a single last peg can end; the colour test (red = 1, 4, 7, 10; blue = 2, 5, 8; green = 3, 6, 9). Mirror-image holes are grouped.

strip	hole	fewest	last peg on	ways	colour test
3	1	1	1	1	says red (1)
3	2	2	—	0	silent (says blue)
4	1, 4	2	—	0	impossible (1,1,1)
4	2 / 3	1	3 / 2	1	says green / blue
5	1, 5	2	—	0	silent
5	2, 4	2	—	0	silent
5	3	2	—	0	impossible (2,2,0)
6	1, 6	3	—	0	silent
6	2, 5	1	2, 5	4	says blue
6	3, 4	2	—	0	silent
7	1, 4, 7	3	—	0	impossible (2,2,2)
7	2, 3, 5, 6	2	—	0	silent
8	1, 8	4	—	0	silent
8	2, 5	1	4, 7	10	says red (1 never occurs)
8	3, 6	3	—	0	impossible (3,3,1)
8	4, 7	1	2, 5	10	says blue (8 never occurs)
9	1, 9	4	—	0	silent
9	2 to 8	2	—	0	silent
10	1, 10	5	—	0	impossible (3,3,3)
10	2, 5 / 6, 9	1	6, 9 / 2, 5	28	says green / blue
10	3, 8	3	—	0	silent
10	4, 7	2	—	0	impossible (3,3,3)

A block of n pegs in the middle of a long empty strip: fewest left 1, 1, 2, 2, 3, 3, 4, 4, 5, 5, 6, 6 for $n = 1, \dots, 12$.

Small boards with one hole.

board	hole	fewest	last peg on	note
2×3	corner / middle of a row	4 / 5	—	one move / no move
2×4	any	5 or 6	—	at most two moves
3×3	centre / edge middle / corner	8 / 7 / 4	—	no move / one move / 14 complete games, ending with 4 (ten of them) or 5
3×4	corner A	1	A, D	852 winning games, 10 jumps
3×4	B (top middle)	1	B only	488 winning games
3×4	E / F (middle row)	10 / 9	—	one / two moves at most
3×5	C (middle of a long side)	1	C only	17,424 winning games, 13 jumps
3×5	other top or bottom cells	2	—	
3×5	middle row	12	—	
4×4	edge, not corner (B)	1	C, O	210,422 winning games, 14 jumps
4×4	corners and the four centre cells	2	—	both diagonal colourings needed
plus (5 cells)	centre / arm	4 / 3	—	no move / one move
cross (9 cells)	any	2	—	
ring of n	any	1 for $n = 3, 4,$ 6, 8, 10, 12; 2 for $n = 5, 7,$ 9, 11	—	4, 8, 40, 112, 336 winning games for $n = 4, 6, 8, 10, 12$

Soldiers: golden weights σ^d for $d = 0, \dots, 9$: 1.000, 0.618, 0.382, 0.236, 0.146, 0.090, 0.056, 0.034, 0.021, 0.013. Exact forms: $\sigma^2 = 1 - \sigma$, $\sigma^3 = 2\sigma - 1$, $\sigma^4 = 2 - 3\sigma$, $\sigma^5 = 5\sigma - 3$. **Weight of the m heaviest cells below the line (target row L):** $L = 1$: 0.618, 1.000 ($m = 1, 2$). $L = 2$: 0.382, 0.618, 0.854, 1.090 ($m = 1..4$). $L = 3$: 0.236, 0.382, 0.528, 0.674, $\dots$, 0.944 ($m = 7$), 1.034 ($m = 8$). $L = 4$: 0.936 ($m = 16$), 0.979 (18), 1.000 (19), 1.021 (20). $L = 5$: the whole half-plane, 1.000.

The twenty-soldier army for row 4, in the coordinates of the 4–5 board (columns a–i, rows 1–8 from the bottom, the line between rows 4 and 5, the star on e8): a4–i4 (nine), d3, e3, f3, g3, d2, e2, f2, g2, h2, e1, f1. Jumps: e3>e4>e5, e1>e2>e3, c4>d4>e4, a4>b4>c4, d2>d3>d4, e4>e5>e6, f3>f4>f5, f1>f2>f3, c4>d4>e4, e3>e4>e5, h2>g2>f2, f2>f3>f4, f4>f5>f6, g3>g4>g5, i4>h4>g4, g4>g5>g6, e5>e6>e7, g6>f6>e6, e6>e7>e8. (Checked by replaying; the last position is a single soldier on e8.)

8	.	.	.	.	4★	.	.	.	.
7	.	.	.	.	.	.	.	.	.
6	.	.	.	.	.	.	.	.	.
5	.	.	.	.	.	.	.	.	.
4	.	.	.	.	.	.	.	.	.
3	.	.	.	.	.	.	.	.	.
2	.	.	.	.	.	.	.	.	.
1	.	.	.	.	.	.	.	.	.
	a	b	c	d	e	f	g	h	i